

Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting

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ICML 2025

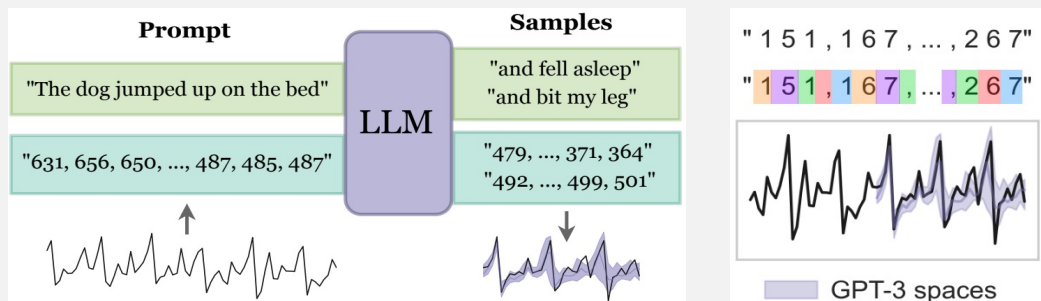
June 2025

- **Time series forecasting (TSF)** plays a pivotal role across numerous domains.
- Traditional deep learning models (LSTM, RNN, Transformer) are widely used but limited in generalizing across diverse time series domains.
- Recent advancements have explored augmenting models with text or vision modalities to improve accuracy.
 - **Text:** provide contextual understanding
 - **Vision:** captures intricate temporal patterns

- Recent methods can be divided into two paradigms.

No-tuning based (LLMTime)

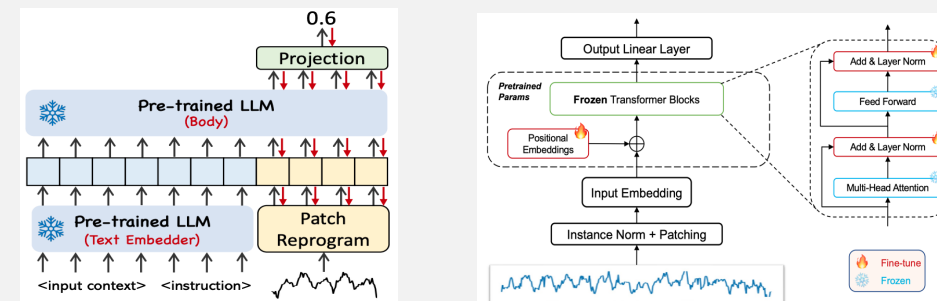
directly encode numerical data as text, use pre-trained LLMs without fine-tuning or training.



Gruver, Nate, et al. “Large language models are zero-shot time series forecasters.” NeurIPS (2024).

Tuning based (Time-LLM, GPT4TS)

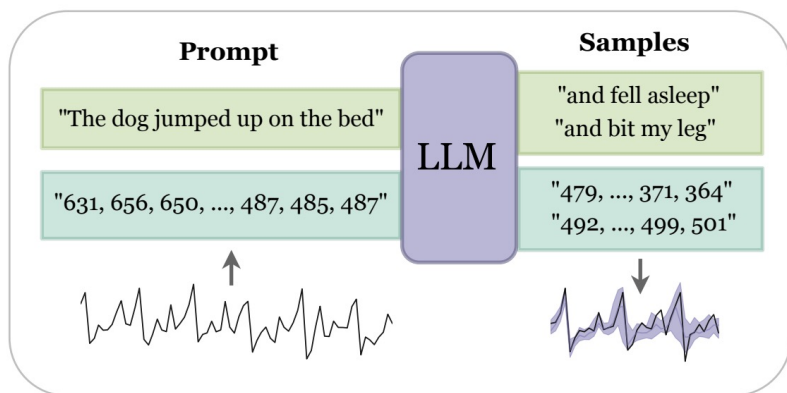
align time series with LLM embeddings, need fine-tuning adaptation layer or LLM itself.



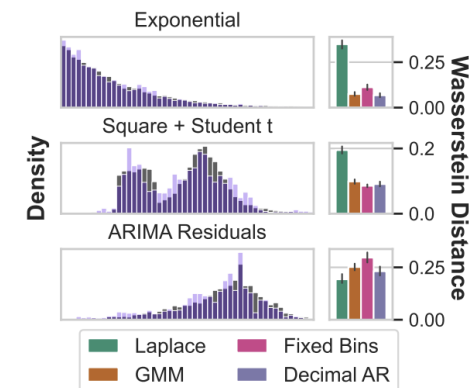
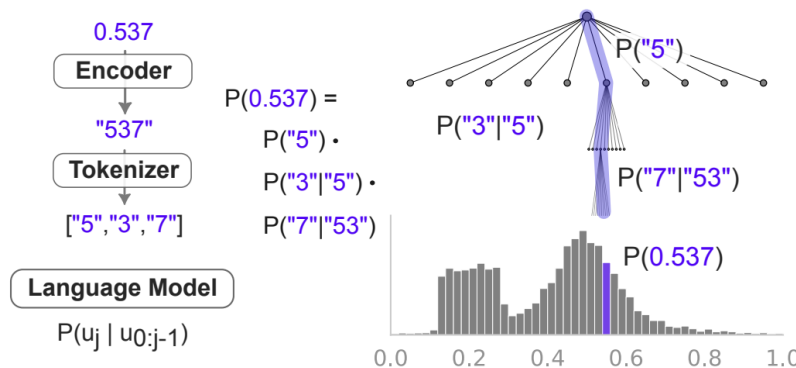
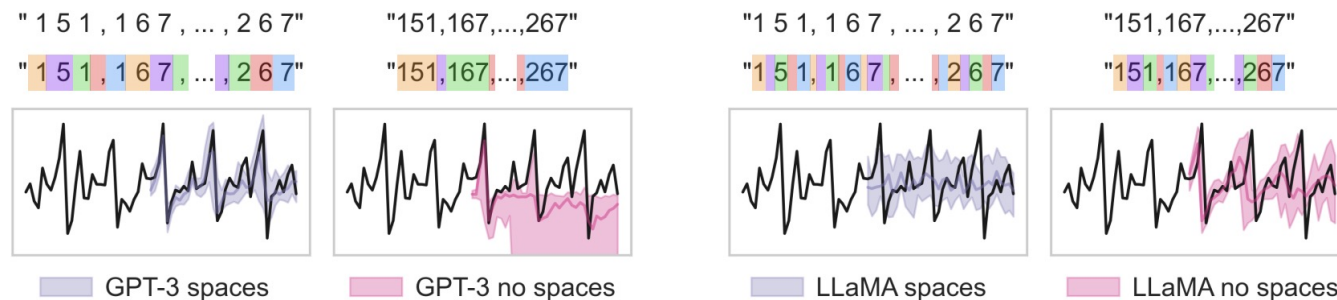
Jin, Ming, et al. "Time-llm: Time series forecasting by reprogramming large language models." ICLR (2024).
Zhou, Tian, et al. "One fits all: Power general time series analysis by pretrained lm." NeurIPS (2023)

Related Work — Text-Augmented Models

- **No-tuning based method:** methods like LLMTime directly encode numerical data as text, use pre-trained LLMs without fine-tuning or training.

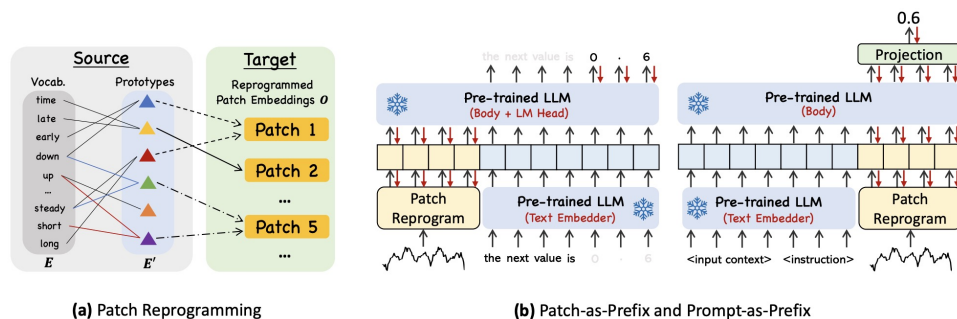
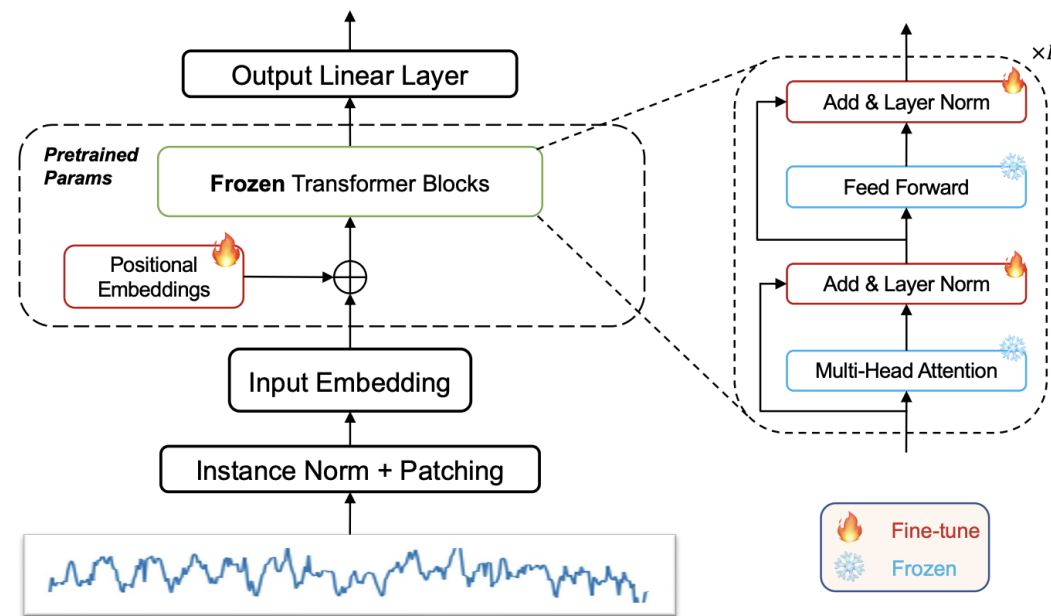
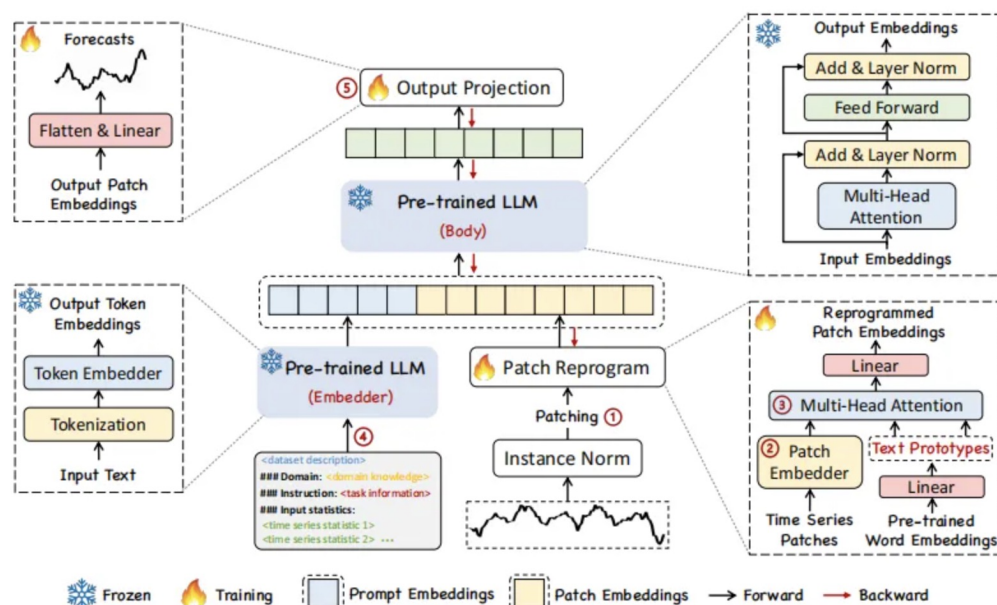


1. LLM's restricted context windows
2. LLM's weakness in arithmetic and recursive operations



Related Work — Text-Augmented Models

- **Tuning based method:** methods like Time-LLM and GPT4TS align time series with LLM embeddings, need fine-tuning adaptation layer or LLM itself.



1. Modality gap between time series and text
2. Pre-trained word-embeddings are rare for TSF

- Visual data is more similar to time series and can well reflect its characteristics.
- With the development of CV, researchers have tried to transform time series into visual representations, enables vision models to identify and exploit underlying patterns.

	Characteristics	Origin	Information
Time series	continuous	physical systems	high redundancy
Image	continuous	physical systems	high redundancy
Text	discrete	human cognitive construct	semantically dense

Related Work — Vision-Augmented Models

- Early approaches explore converting time series into different types of images, and then using a visual model for feature extraction and prediction.

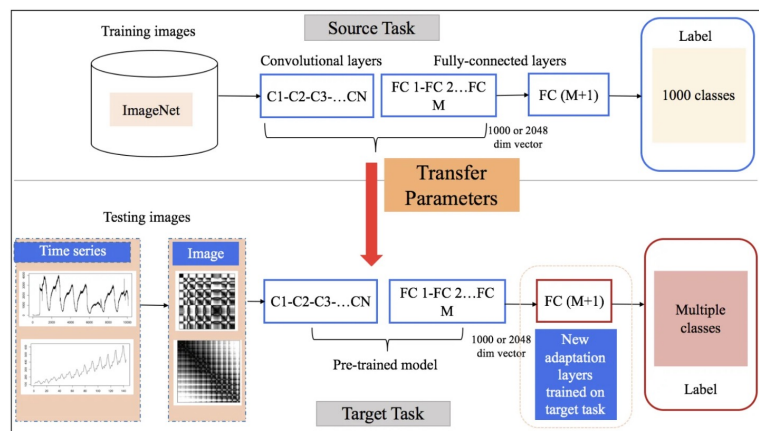
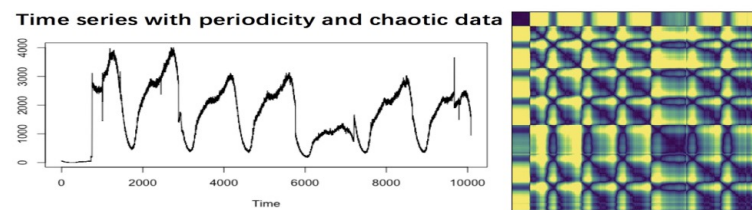


Image: Recurrence plot

Vision model: Convolutional Neural Network

Loss function: OWA (MASE + sMAPE)

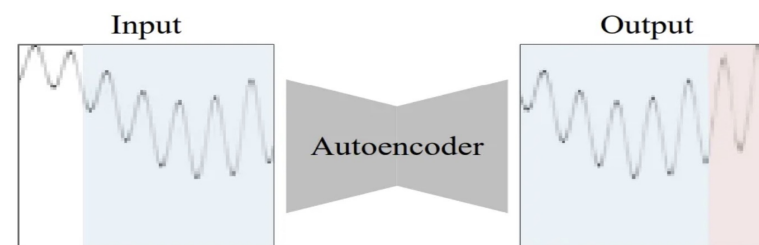
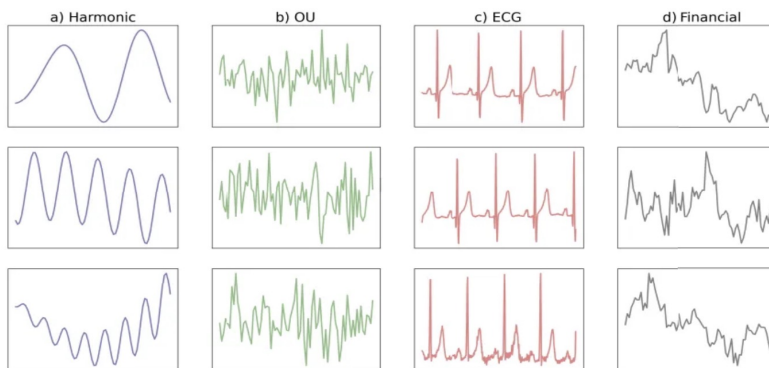


Image: X-time points, Y-normalized values.

Vision model: Convolutional Autoencoder

Loss function: Jensen-Shannon divergence

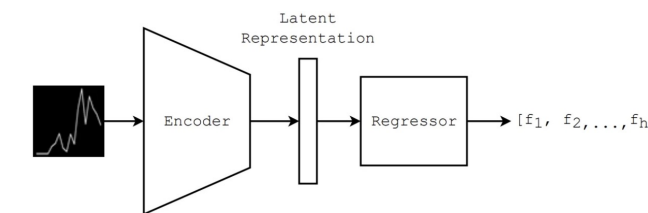
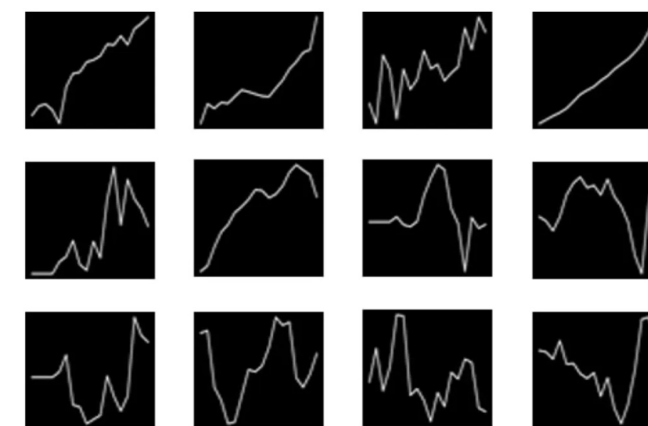


Image: Line graph

Vision model: Convolutional Neural Network

Loss function: MSE

Related Work — Vision-Augmented Models

- **TimeNet** advances the field with multi-scale frequency-based time image transformations.

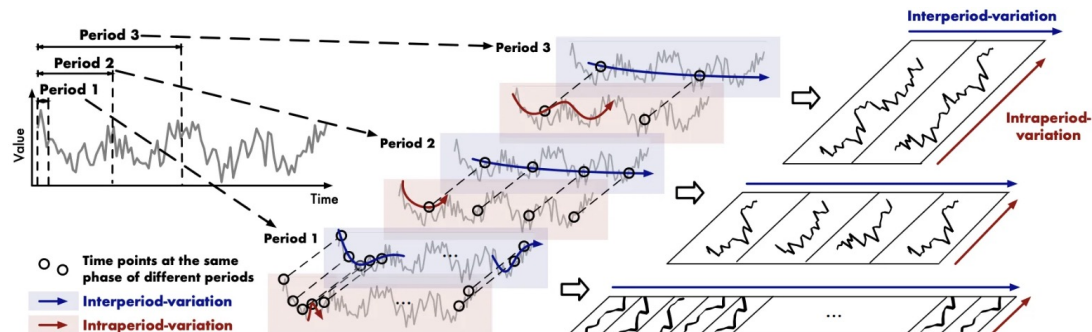


Figure 1: Multi-periodicity and temporal 2D-variation of time series. Each period involves the **intraperiod-variation** and **interperiod-variation**. We transform the original 1D time series into a set of 2D tensors based on multiple periods, which can unify the intraperiod- and interperiod-variations.

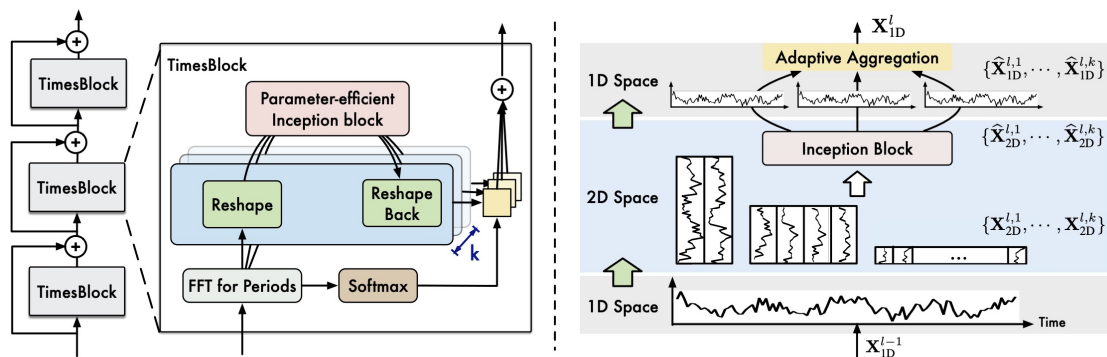


Figure 3: Overall architecture of TimesNet. TimesNet is stacked by TimesBlocks in a residual way. TimesBlocks can capture various temporal 2D-variations from k different reshaped tensors by a parameter-efficient inception block in 2D space and fuse them based on normalized amplitude values.

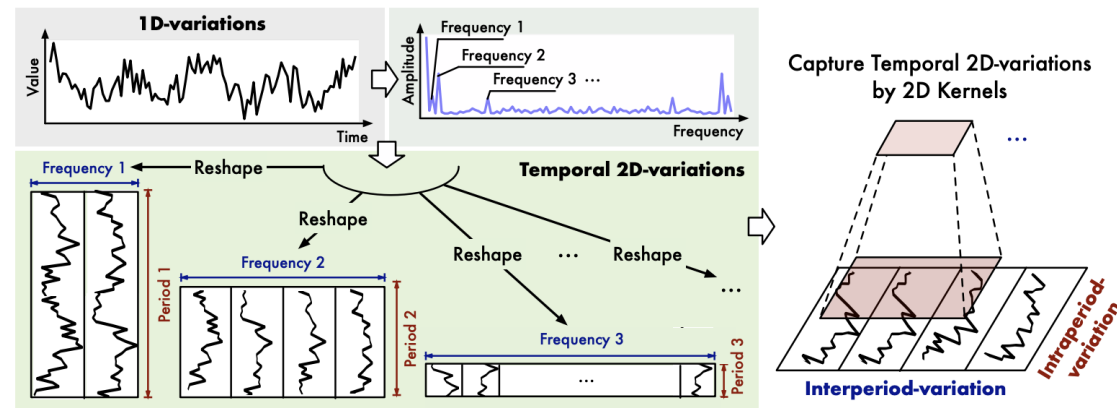


Figure 2: A univariate example to illustrate 2D structure in time series. By discovering the periodicity, we can transform the original 1D time series into structured 2D tensors, which can be processed by 2D kernels conveniently. By conducting the same reshape operation to all variates of time series, we can extend the above process to multivariate time series.

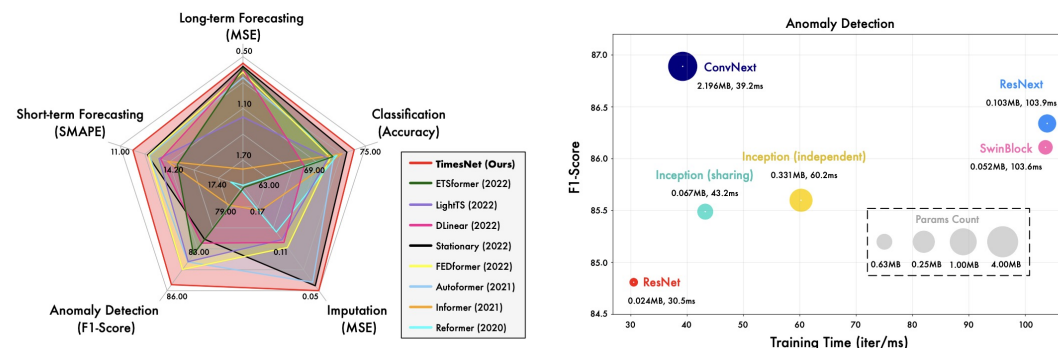
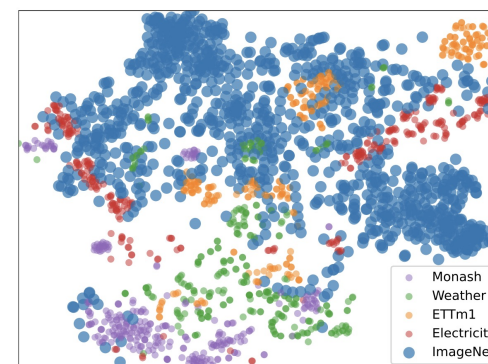
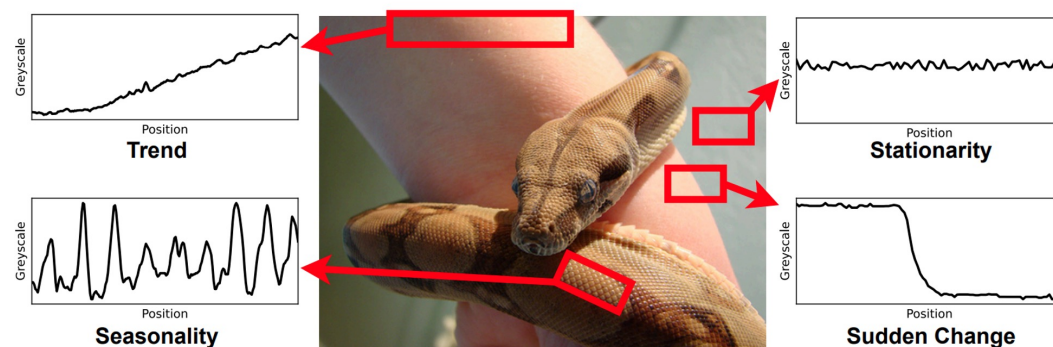
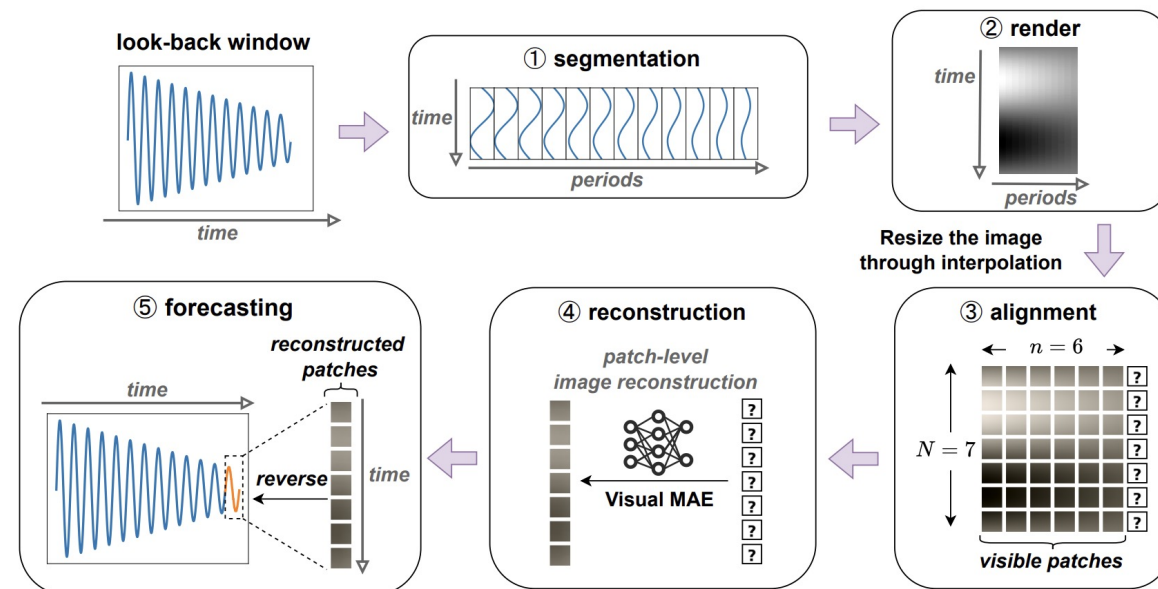
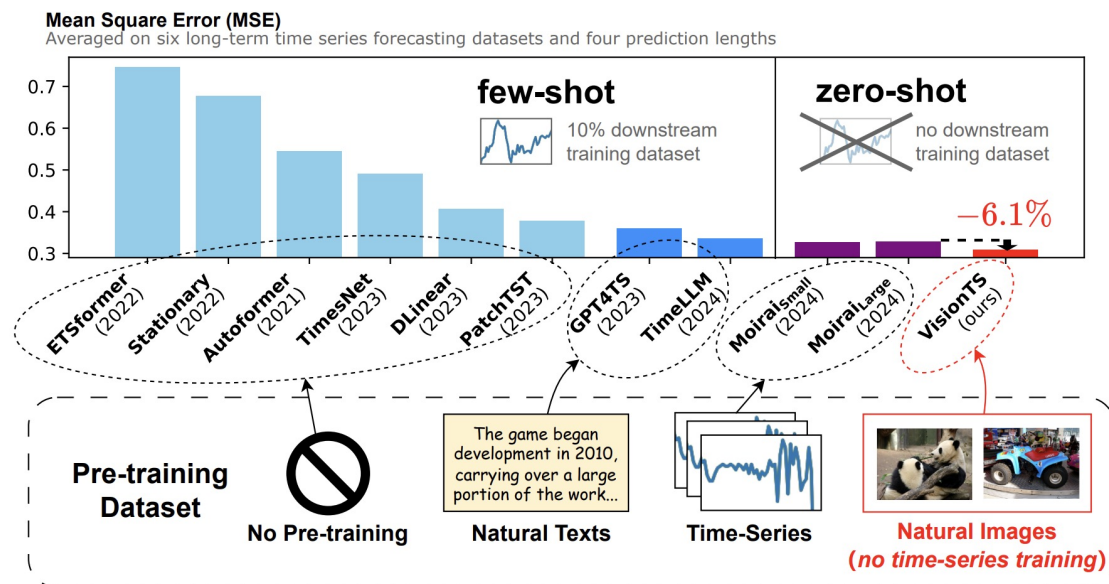


Figure 4: Model performance comparison (left) and generality in different vision backbones (right).

Related Work — Vision-Augmented Models

- **VisionTS** pioneers pre-trained visual encoders with grayscale time series images.



- **Time-VLM** (Time-**V**ision-**L**anguage **M**odel)
- Can we bridge temporal, visual, and textual modalities for enhanced TSF?

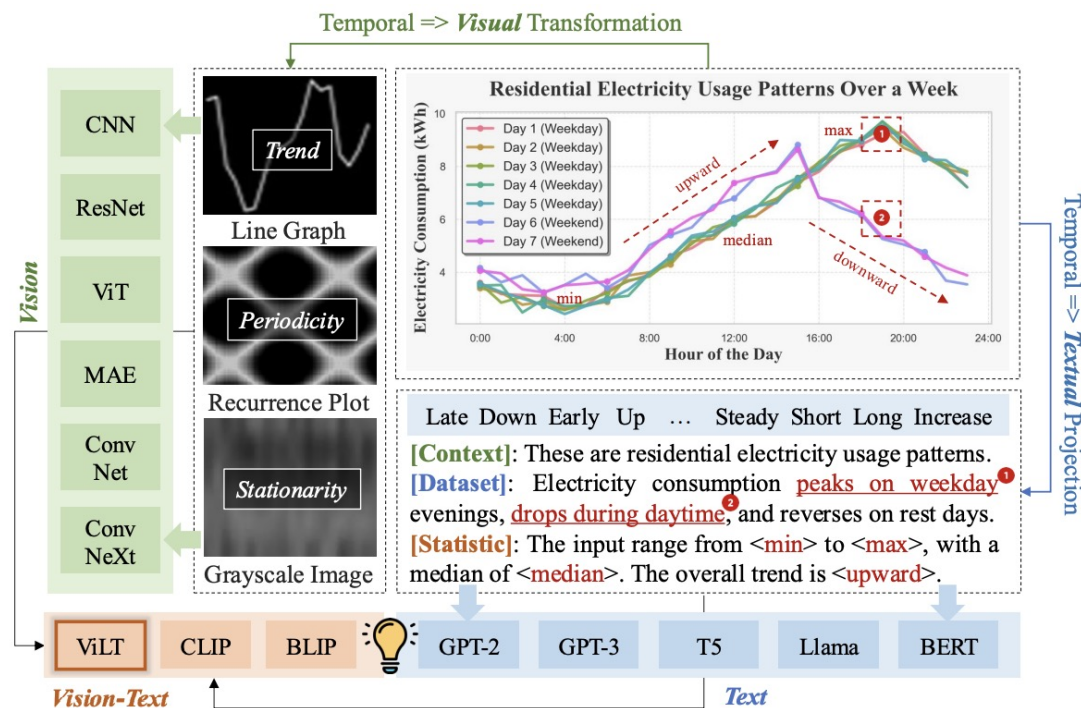


Figure 1: Our Time-VLM combines text (Right) and vision (Left) modalities to augment time series forecasting.

Methodology

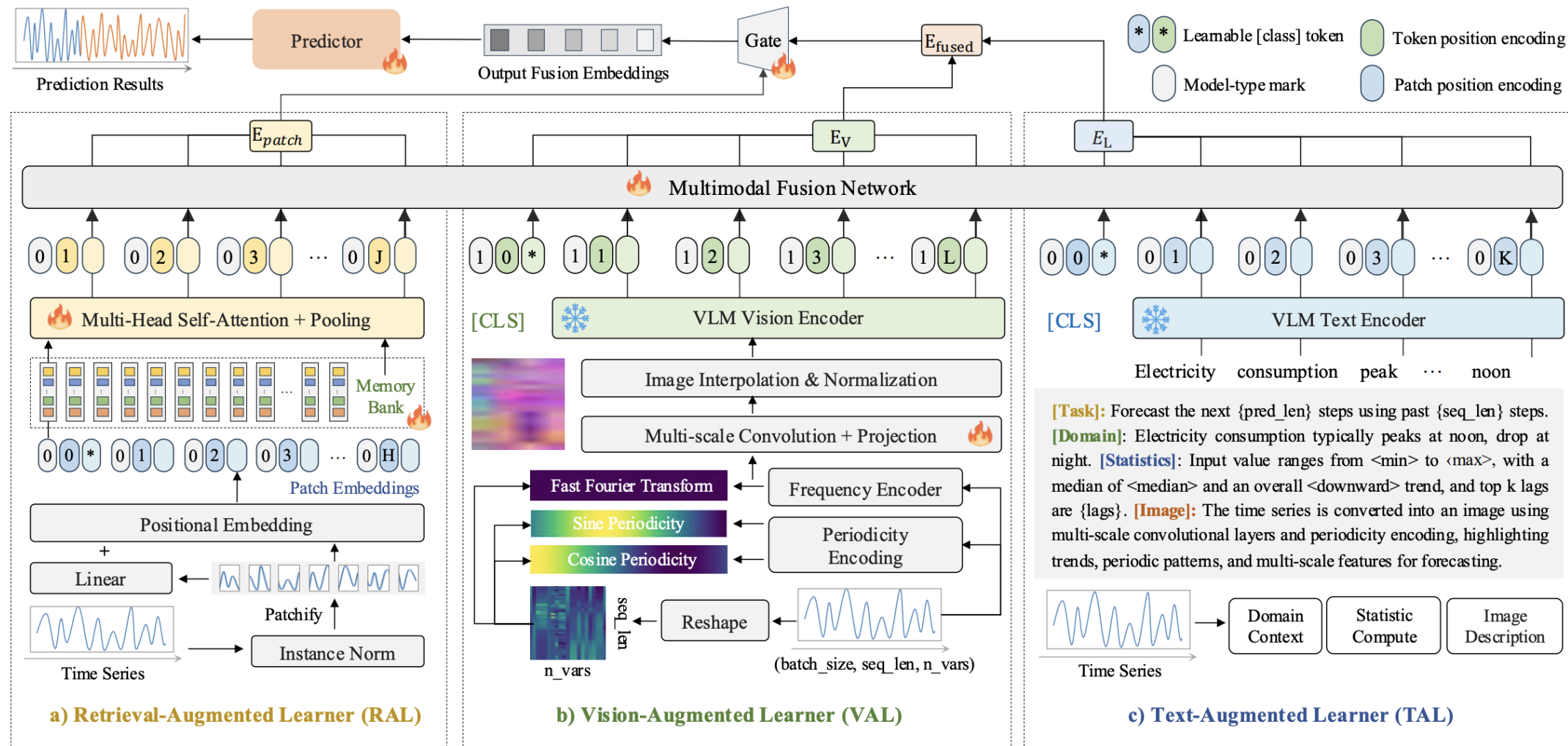


Figure 2: Overview of the Time-VLM framework.

- Dataset Details

Table 8: Summary of benchmark datasets. Each dataset includes multiple time series (Dim.) with varying sequence lengths, split into training, validation, and testing sets. Data are collected at different frequencies across various domains.

Tasks	Dataset	Dim.	Series Length	Dataset Size	Frequency	Domain	Periodicity
Long-term Forecasting	ETTm1	7	{96, 192, 336, 720}	(34465, 11521, 11521)	15 min	Temperature	96
	ETTm2	7	{96, 192, 336, 720}	(34465, 11521, 11521)	15 min	Temperature	96
	ETTTh1	7	{96, 192, 336, 720}	(8545, 2881, 2881)	1 hour	Temperature	24
	ETTTh2	7	{96, 192, 336, 720}	(8545, 2881, 2881)	1 hour	Temperature	24
	Electricity	321	{96, 192, 336, 720}	(18317, 2633, 5261)	1 hour	Electricity	24
	Traffic	862	{96, 192, 336, 720}	(12185, 1757, 3509)	1 hour	Transportation	24
	Weather	21	{96, 192, 336, 720}	(36792, 5271, 10540)	10 min	Weather	144
Short-term Forecasting	M4 - Yearly	1	6	(23000, 0, 23000)	Yearly	Demographic	1
	M4 - Quarterly	1	8	(24000, 0, 24000)	Quarterly	Finance	4
	M4 - Monthly	1	18	(48000, 0, 48000)	Monthly	Industry	3
	M4 - Weekly	1	13	(359, 0, 359)	Weekly	Macro	4
	M4 - Daily	1	14	(4227, 0, 4227)	Daily	Micro	1
	M4 - Hourly	1	48	(414, 0, 414)	Hourly	Other	24

- Optimization Settings

Table 9: Default Model Architecture Parameters

Parameter	Default Value	Description
image_size	64	Size of generated image representation
d_model	128	Dimension of hidden embeddings
d_fusion	256	Dimension of gated fusion module
num_workers	32	Number of data loader workers
e_layers	2	Number of encoder layers
d_layers	1	Number of decoder layers
dropout	0.1	Dropout rate
vlm_fused_len	156	Token length of VLM multimodal embedding
vlm_hidden_dim	768	Hidden dimension of VLM

- Evaluation Metrics

$$\text{MSE} = \frac{1}{H} \sum_{h=1}^T (\mathbf{Y}_h - \hat{\mathbf{Y}}_h)^2,$$

$$\text{SMAPE} = \frac{200}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h| + |\hat{\mathbf{Y}}_h|},$$

$$\text{MASE} = \frac{1}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{\frac{1}{H-s} \sum_{j=s+1}^H |\mathbf{Y}_j - \mathbf{Y}_{j-s}|},$$

$$\text{MAE} = \frac{1}{H} \sum_{h=1}^H |\mathbf{Y}_h - \hat{\mathbf{Y}}_h|,$$

$$\text{MAPE} = \frac{100}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h|},$$

$$\text{OWA} = \frac{1}{2} \left[\frac{\text{SMAPE}}{\text{SMAPE}_{\text{Naive2}}} + \frac{\text{MASE}}{\text{MASE}_{\text{Naive2}}} \right],$$

Table 10: Default Training Parameters

Parameter	Default Value	Description
batch_size	32	Training batch size
learning_rate	0.001	Initial learning rate
training_epochs	10	Number of training epochs
patience	3	Early stopping patience
loss	MSE	Mean square error
seq_len	512	Input sequence length
c_out	7 (ETTh1/h2/m1/m2) 21 (Weather) 321 (Electricity) 862 (Traffic)	Output dimension (dataset-specific)
pred_len	96/192/336/720	Prediction length
periodicity	24 (ETTh1/h2/Electricity/Traffic) 96 (ETTm1/m2) 144 (Weather)	Dataset periodicity (dataset-specific)
norm_const	0.4	Normalization coefficient
patch_len	16	Patch length
padding	8	Padding length
stride	8	Stride length
num_queries	8	Number of learnable queries for temporal memory
n_heads	4	Number of attention heads

Few-shot learning results

Table 1: Few-shot learning on 5% training data. Results are averaged over forecasting horizons $H \in \{96, 192, 336, 720\}$. Lower values indicate better performance. Full results see Section B.1. **Red**: best, **Blue**: second best.

Methods	Time-VLM _{143M} (Ours)		Time-LLM _{3405M} (2024)		GPT4TS (2023)		DLinear (2023)		PatchTST (2023)		TimesNet (2023a)		FEDformer (2022)		Autoformer (2021)		Stationary (2022b)		ETSformer (2022)		LightTS (2022)		Informer (2021)		Reformer (2020)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.442	0.453	0.627	0.543	0.681	0.560	0.750	0.611	0.694	0.569	0.925	0.647	0.658	0.562	0.722	0.598	0.943	0.646	1.189	0.839	1.451	0.903	1.225	0.817	1.241	0.835
ETTh2	0.354	0.402	0.382	0.418	0.400	0.433	0.694	0.577	0.827	0.615	0.439	0.448	0.463	0.454	0.441	0.457	0.470	0.489	0.809	0.681	3.206	1.268	3.922	1.653	3.527	1.472
ETTm1	0.364	0.385	0.425	0.434	0.472	0.450	0.400	0.417	0.526	0.476	0.717	0.561	0.730	0.592	0.796	0.620	0.857	0.598	1.125	0.782	1.123	0.765	1.163	0.791	1.264	0.826
ETTm2	0.262	0.323	0.274	0.323	0.308	0.346	0.399	0.426	0.314	0.352	0.344	0.372	0.381	0.404	0.388	0.433	0.341	0.372	0.534	0.547	1.415	0.871	3.658	1.489	3.581	1.487
Weather	0.240	0.280	0.260	0.309	0.263	0.301	0.263	0.308	0.269	0.303	0.298	0.318	0.309	0.353	0.310	0.353	0.327	0.328	0.333	0.371	0.305	0.345	0.584	0.527	0.447	0.453
ECL	0.218	0.315	0.179	0.268	0.178	0.273	0.176	0.275	0.181	0.277	0.402	0.453	0.266	0.353	0.346	0.404	0.627	0.603	0.800	0.685	0.878	0.725	1.281	0.929	1.289	0.904
Traffic	0.558	0.410	0.423	0.298	0.434	0.305	0.450	0.317	0.418	0.296	0.867	0.493	0.676	0.423	0.833	0.502	1.526	0.839	1.859	0.927	1.557	0.795	1.591	0.832	1.618	0.851

Table 2: Few-shot learning on 10% training data. We use the same protocol in Table 1. Full results see Section B.1.

Methods	Time-VLM _{143M} (Ours)		Time-LLM _{3405M} (2024)		GPT4TS (2023)		DLinear (2023)		PatchTST (2023)		TimesNet (2023a)		FEDformer (2022)		Autoformer (2021)		Stationary (2022b)		ETSformer (2022)		LightTS (2022)		Informer (2021)		Reformer (2020)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.431	0.442	0.556	0.522	0.590	0.525	0.691	0.600	0.633	0.542	0.869	0.628	0.639	0.561	0.702	0.596	0.915	0.639	1.180	0.834	1.375	0.877	1.199	0.809	1.249	0.833
ETTh2	0.361	0.405	0.370	0.394	0.397	0.421	0.605	0.538	0.415	0.431	0.479	0.465	0.466	0.475	0.488	0.499	0.462	0.455	0.894	0.713	2.655	1.160	3.872	1.513	3.485	1.486
ETTm1	0.360	0.382	0.404	0.427	0.464	0.441	0.411	0.429	0.501	0.466	0.677	0.537	0.722	0.605	0.802	0.628	0.797	0.578	0.980	0.714	0.971	0.705	1.192	0.821	1.426	0.856
ETTm2	0.263	0.323	0.277	0.323	0.293	0.335	0.316	0.368	0.296	0.343	0.320	0.353	0.463	0.488	1.342	0.930	0.332	0.366	0.447	0.487	0.987	0.756	3.370	1.440	3.978	1.587
Weather	0.233	0.274	0.234	0.273	0.238	0.275	0.241	0.283	0.242	0.279	0.279	0.301	0.284	0.324	0.300	0.342	0.318	0.323	0.318	0.360	0.289	0.322	0.597	0.495	0.546	0.469
ECL	0.198	0.291	0.175	0.270	0.176	0.269	0.180	0.280	0.180	0.273	0.323	0.392	0.346	0.427	0.431	0.478	0.444	0.480	0.660	0.617	0.441	0.489	1.195	0.891	0.965	0.768
Traffic	0.484	0.357	0.429	0.306	0.440	0.310	0.447	0.313	0.430	0.305	0.951	0.535	0.663	0.425	0.749	0.446	1.453	0.815	1.914	0.936	1.248	0.684	1.534	0.811	1.551	0.821

Zero-shot learning results

Table 3: Zero-shot learning results. Full results see Section B.2.

Methods	Time-VLM _{143M} (Ours)		Time-LLM _{3405M} (2024)		LLMTime (2023)		GPT4TS (2023)		DLinear (2023)		PatchTST (2023)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1 → ETTh2	0.338	0.385	0.353	0.387	0.992	0.708	0.406	0.422	0.493	0.488	0.380	0.405
ETTh1 → ETTm2	0.293	0.350	0.273	0.340	1.867	0.869	0.325	0.363	0.415	0.452	0.314	0.360
ETTh2 → ETTh1	0.496	0.480	0.479	0.474	1.961	0.981	0.757	0.578	0.703	0.574	0.565	0.513
ETTh2 → ETTm2	0.297	0.353	0.272	0.341	1.867	0.869	0.335	0.370	0.328	0.386	0.325	0.365
ETTm1 → ETTh2	0.354	0.397	0.381	0.412	0.992	0.708	0.433	0.439	0.464	0.475	0.439	0.438
ETTm1 → ETTm2	0.264	0.319	0.268	0.320	1.867	0.869	0.313	0.348	0.335	0.389	0.296	0.334
ETTh2 → ETTm2	0.359	0.399	0.354	0.400	0.992	0.708	0.435	0.443	0.455	0.471	0.409	0.425
ETTm2 → ETTm1	0.432	0.426	0.414	0.438	1.933	0.984	0.769	0.567	0.649	0.537	0.568	0.492

- Full-shot learning results

Table 4: Short-term time series forecasting results on M4. The forecasting horizons are in [6, 48] and the three rows provided are weighted averaged from all datasets under different sampling intervals. Full results see [Section B.3](#).

Methods	Time-VLM _{143M} (Ours)	Time-LLM _{3405M} (2024)	GPT4TS (2023)	TimesNet (2023a)	PatchTST (2023)	N-HiTS (2023)	N-BEATS (2020)	ETSformer (2022)	LightTS (2022)	DLinear (2023)	FEDformer (2022)	Stationary (2022b)	Autoformer (2021)	Informer (2021)	Reformer (2020)
Average SMAPE	11.894	<u>11.983</u>	12.690	12.880	12.059	12.035	12.250	14.718	13.525	13.639	13.160	12.780	12.909	14.086	18.200
Average MASE	1.592	<u>1.595</u>	1.808	1.836	1.623	1.625	1.698	2.408	2.111	2.095	1.775	1.756	1.771	2.718	4.223
Average OWA	0.855	<u>0.859</u>	0.940	0.955	0.869	0.869	0.896	1.172	1.051	1.051	0.949	0.930	0.939	1.230	1.775

Table 5: Long-term forecasting results. We use the same protocol in [Table 1](#). Full results see in [Section B.4](#).

Methods	Time-VLM _{143M} (Ours)		Time-LLM _{3405M} (2024)		GPT4TS (2023)		DLinear (2023)		PatchTST (2023)		TimesNet (2023a)		FEDformer (2022)		Autoformer (2021)		Stationary (2022b)		ETSformer (2022)		LightTS (2022)		Informer (2021)		Reformer (2020)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.405	0.420	<u>0.408</u>	<u>0.423</u>	0.465	0.455	0.422	0.437	0.413	0.430	0.458	0.450	0.440	0.460	0.496	0.487	0.570	0.537	0.542	0.510	0.491	0.479	1.040	0.795	1.029	0.805
ETTh2	0.341	0.391	<u>0.334</u>	<u>0.383</u>	0.381	0.412	0.431	0.446	0.330	0.379	0.414	0.427	0.437	0.449	0.450	0.459	0.526	0.516	0.439	0.452	0.602	0.543	4.431	1.729	6.736	2.191
ETTm1	<u>0.347</u>	<u>0.377</u>	0.329	0.372	0.388	0.403	0.357	0.378	0.351	0.380	0.400	0.406	0.448	0.452	0.588	0.517	0.481	0.456	0.429	0.425	0.435	0.437	0.961	0.734	0.799	0.671
ETTm2	0.248	0.311	<u>0.251</u>	<u>0.313</u>	0.284	0.339	0.267	0.333	0.255	0.315	0.291	0.333	0.305	0.349	0.327	0.371	0.306	0.347	0.293	0.342	0.409	0.436	1.410	0.810	1.479	0.915
Weather	0.224	<u>0.263</u>	<u>0.225</u>	0.257	0.237	0.270	0.248	0.300	0.225	0.264	0.259	0.287	0.309	0.360	0.338	0.382	0.288	0.314	0.271	0.334	0.261	0.312	0.634	0.548	0.803	0.656
Electricity	0.172	0.273	0.158	0.252	0.167	<u>0.263</u>	0.166	<u>0.263</u>	<u>0.161</u>	0.252	0.192	0.295	0.214	0.327	0.227	0.338	0.193	0.296	0.208	0.323	0.229	0.329	0.311	0.397	0.338	0.422
Traffic	0.419	0.303	0.388	<u>0.264</u>	0.414	0.294	0.433	0.295	<u>0.390</u>	0.263	0.620	0.336	0.610	0.376	0.628	0.379	0.624	0.340	0.621	0.396	0.622	0.392	0.764	0.416	0.741	0.422

• Ablation Studies

Table 6: Ablation study on multimodal components.

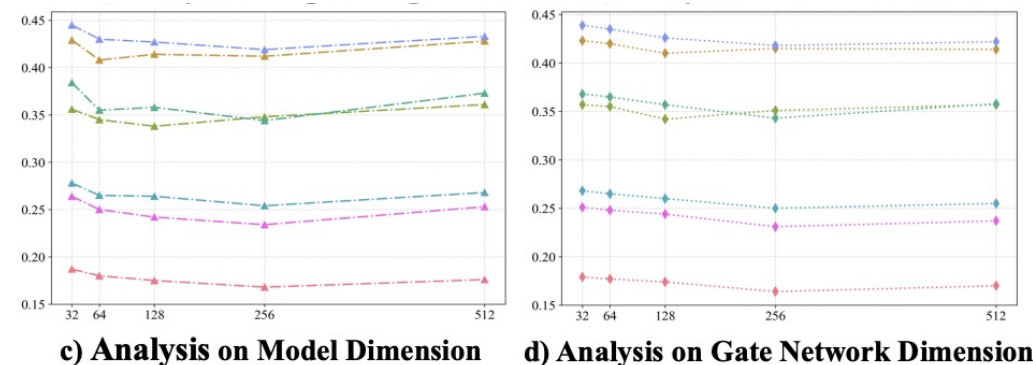
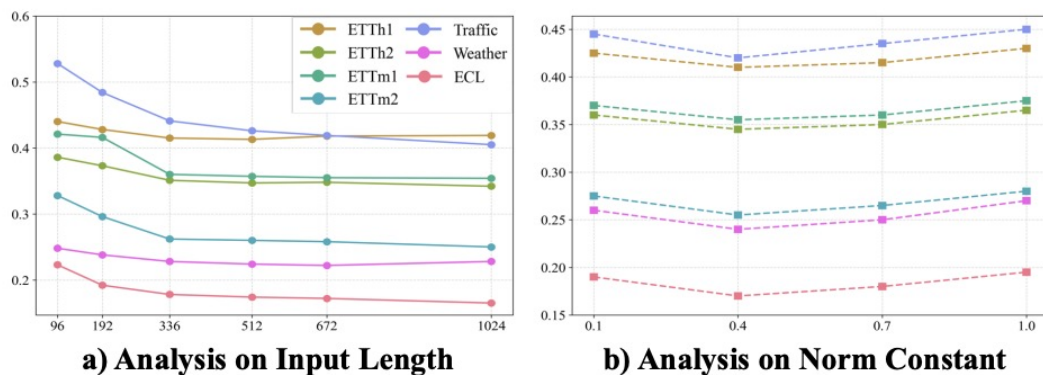
Horizon	Full		w/o RAL		w/o VAL		w/o TAL	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
96	0.160	0.213	0.273	0.324	0.213	0.266	<u>0.165</u>	<u>0.218</u>
192	0.203	0.252	0.297	0.338	0.237	0.298	<u>0.208</u>	<u>0.257</u>
336	0.253	0.291	0.325	0.354	0.255	0.302	<u>0.258</u>	<u>0.295</u>
720	0.317	0.340	0.369	0.383	0.309	0.357	<u>0.322</u>	<u>0.345</u>
Avg	0.233	0.274	0.316	0.350	0.254	0.306	<u>0.238</u>	<u>0.279</u>
%Deg	—	—	35.6% ↑	27.7% ↑	9.0% ↑	11.7% ↑	2.1% ↑	1.8% ↑

• Computation Studies

Table 7: Computational efficiency comparison between Time-VLM and Time-LLM across datasets. “-” denotes memory exceeds 49GB, infeasible on a single GPU. Results are averaged over multiple prediction steps under consistent conditions.

Method	Metric	ETTh1	ETTh2	ETTm1	ETTm2	Weather	ECL	Traffic
Time-VLM	Param. (M)	143.6	143.6	143.6	143.6	143.6	143.6	143.6
	Mem. (MiB)	2630	2630	2640	2640	1968	10818	24916
	Speed (s/iter)	0.481	0.438	0.277	0.210	0.296	0.206	0.323
Time-LLM	Param. (M)	<u>3404.6</u>	<u>3404.6</u>	<u>3404.6</u>	<u>3404.6</u>	-	-	-
	Mem. (MiB)	<u>37723</u>	<u>37723</u>	<u>37849</u>	<u>37849</u>	-	-	-
	Speed (s/iter)	<u>0.607</u>	<u>0.553</u>	<u>0.349</u>	<u>0.265</u>	-	-	-

• Hyperparameter Studies



- Multimodal and Few/Zero-shot Analysis

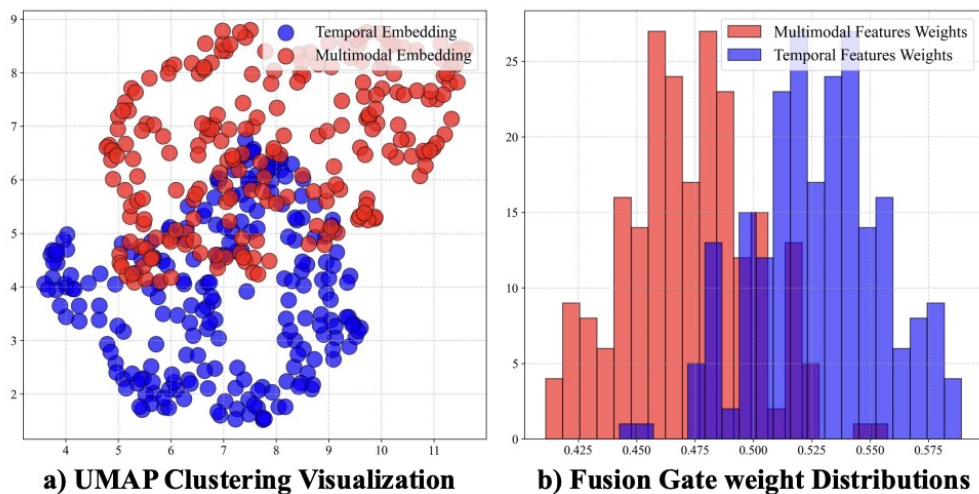


Figure 3: 2D UMAP visualization (Left) and Gate weight distributions (Right) of multimodal and temporal memory embeddings, highlighting their complementary behavior.

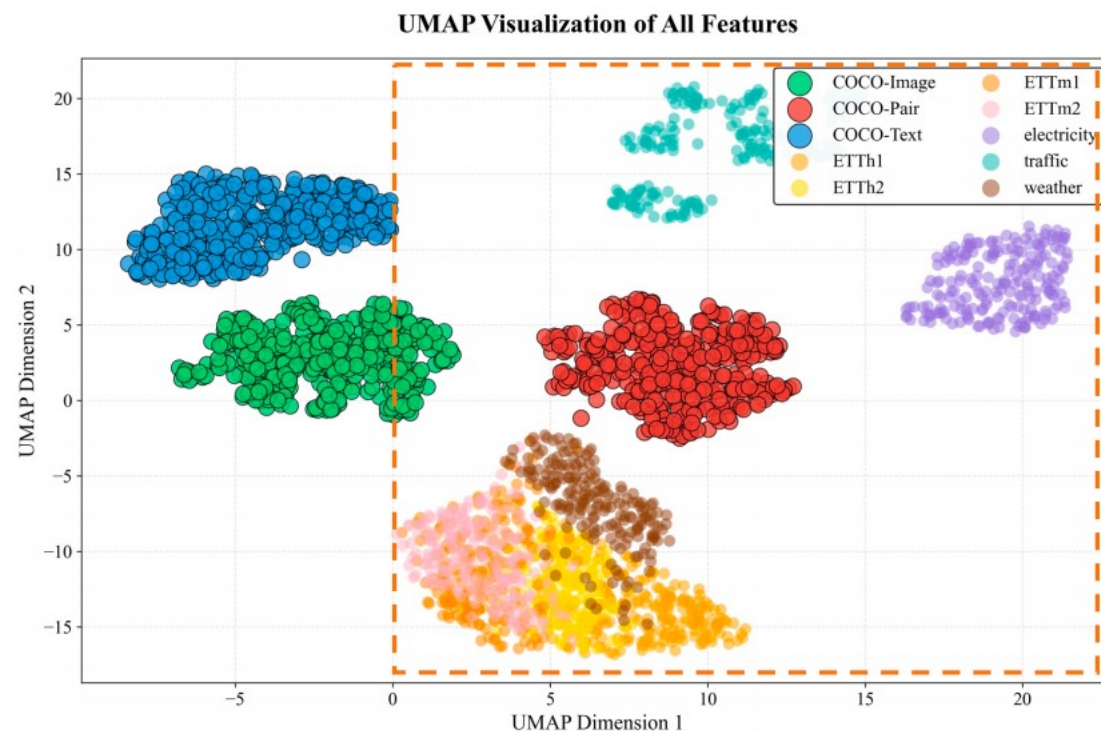


Figure 4: Interpretability visualization of Time-VLM: multimodal feature alignment via UMAP.

- Visualization of Generated Time Series Images



Figure 5: Time series transformed images, capturing key temporal characteristics, including trends, stationarity, seasonality, sudden changes, and frequency-domain patterns.

- Visualization of prediction results

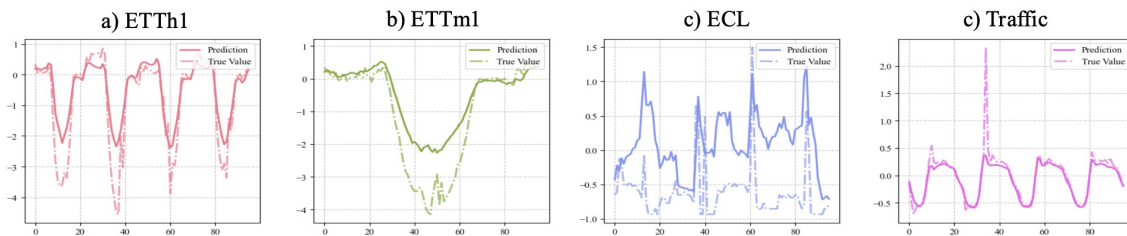


Figure 6: Prediction results visualization for ETTh1, ETTm1, ECL, and Traffic datasets at 96 prediction lengths. True values (solid line) and predicted values (dashed line) are shown for each dataset and horizon.

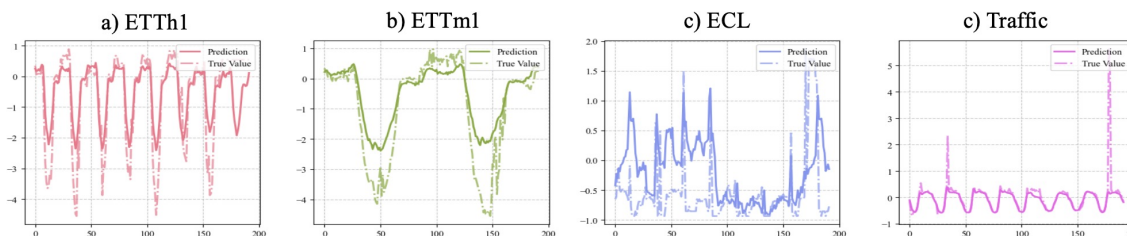


Figure 7: Prediction results visualization for ETTh1, ETTm1, ECL, and Traffic datasets at 192 prediction lengths. True values (solid line) and predicted values (dashed line) are shown for each dataset and horizon.

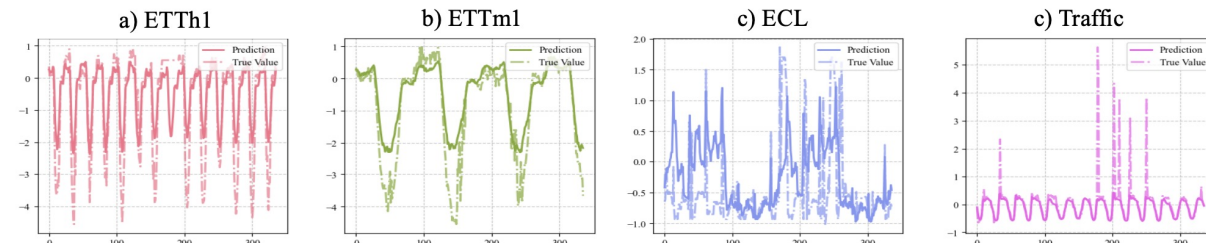


Figure 8: Prediction results visualization for ETTh1, ETTm1, ECL, and Traffic datasets at 336 prediction lengths. True values (solid line) and predicted values (dashed line) are shown for each dataset and horizon.

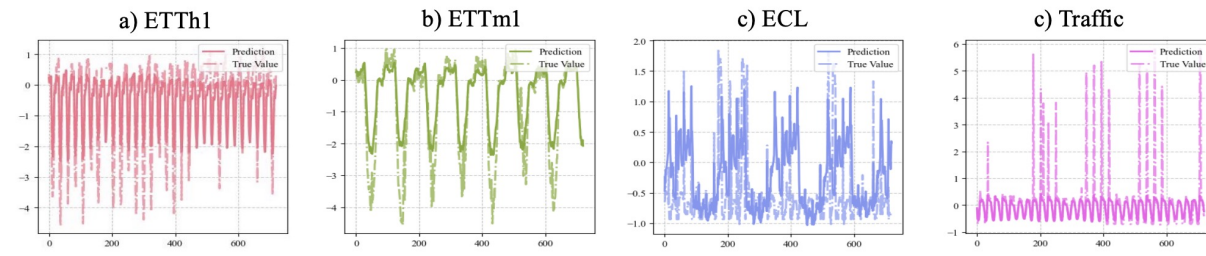


Figure 9: Prediction results visualization for ETTh1, ETTm1, ECL, and Traffic datasets at 720 prediction lengths. True values (solid line) and predicted values (dashed line) are shown for each dataset and horizon.

- Time-VLM is a novel framework leveraging pretrained VLMs to unify temporal, visual, and textual modalities for time series forecasting.
- It is self-enhanced, operating solely on original time series data without external information. By integrating the RAL, VAL, and TAL, Time-VLM bridges modality gaps, enabling rich cross-modal interactions.

- Future Work:
 - High quality multi-modal time series benchmark => foundation model for multi-tasks.
 - Time series imaging — solving the problem of information density misalignment when different input step lengths are mapped to fixed resolution images.
 - Visual distillation — visual encoder may be redundant compared to time series.
 - Enhanced text encoder with time series semantic understanding capabilities.
 - Interpretability, modal contribution and analysis.

Thanks for listening!



June 2025